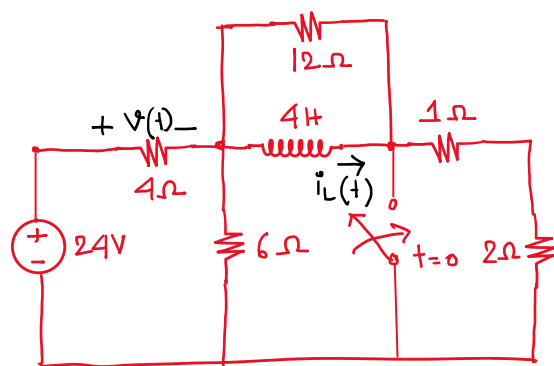
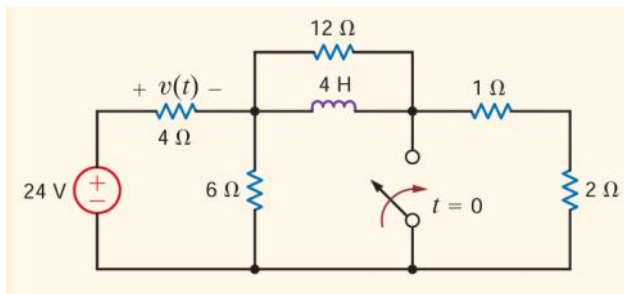
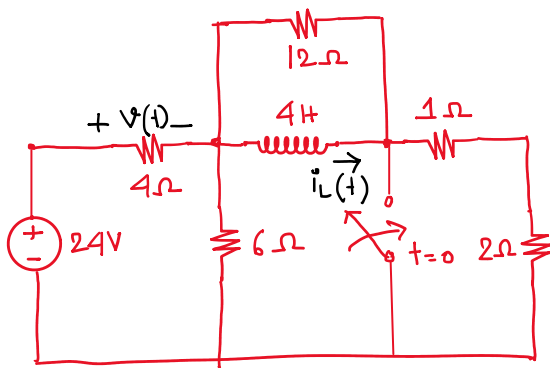


RL circuit forced response

Sunday, January 4, 2026 6:15 PM



$-\infty$

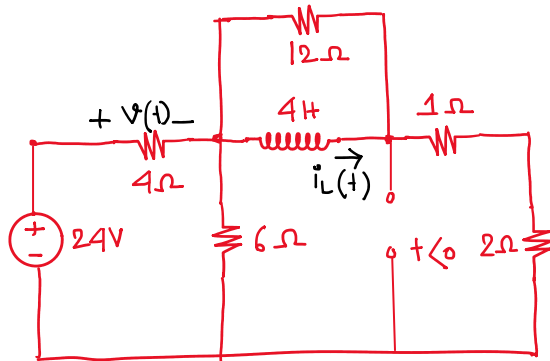


$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau}$$

steps: determine $i_L(\infty)$, $i_L(0)$, τ

$i_L(0)$: for $t < 0$; Switch open

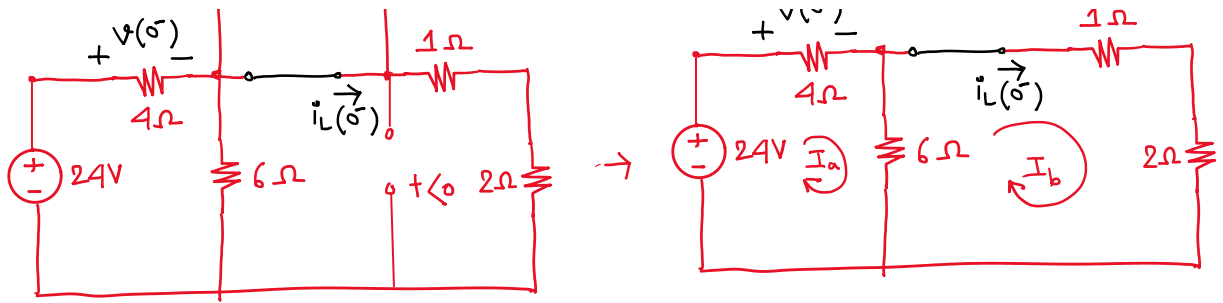
$$i_L(0) = i_L(0^-) = i_L(0^+)$$



$t = 0^-$: [Inductor shorted to DC source]

12Ω shorted.





Mesh Analysis: Loop a: $24 - 4I_a - 6(I_a - I_b) = 0$ —(i)

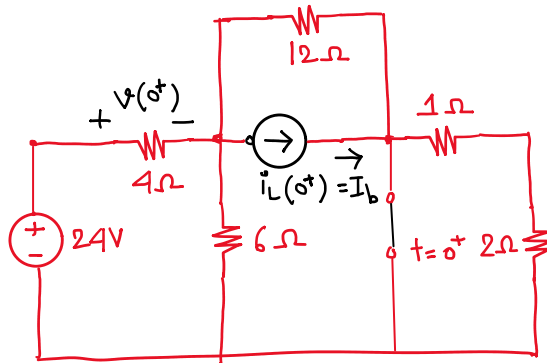
Loop b: $-6(I_b - I_a) - 3I_b = 0$ —(ii)

Solving (i) & (ii) we find I_b & $i_L(0^-) = I_b$

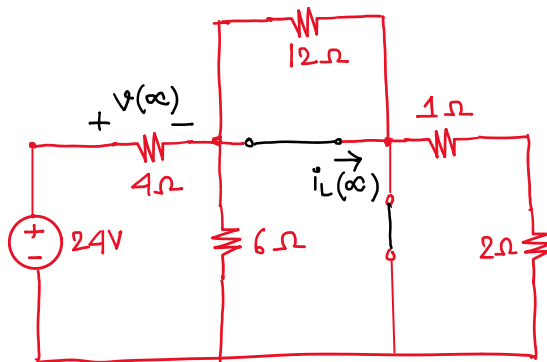
$i_L(0) = i_L(0^-) = i_L(0^+) = I_b$

Step 2: determine $i_L(\infty)$

Q $t=0^+$, switch is closed
and $i_L(0^+) = i_L(0^-)$ [Current through inductor doesn't change instantaneously]
So, $i_L(0^+) = i_L(0^-) = I_b$
at $t=0^+$, we can consider that the inductor is current source with $i_L(0^+) = I_b$

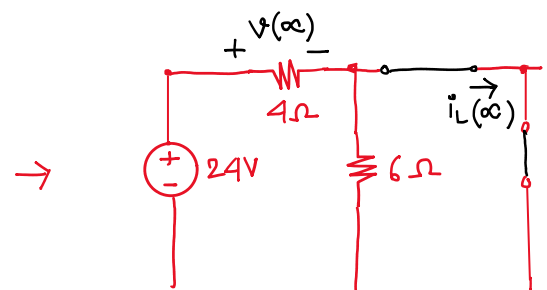


at $t=\infty$ [Steady state condition. inductor short ckt]



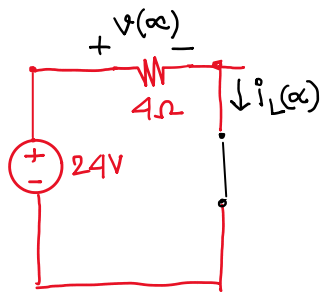
6Ω Resistor shorted

12Ω Resistor and (1+2)Ω Resistor are shorted.



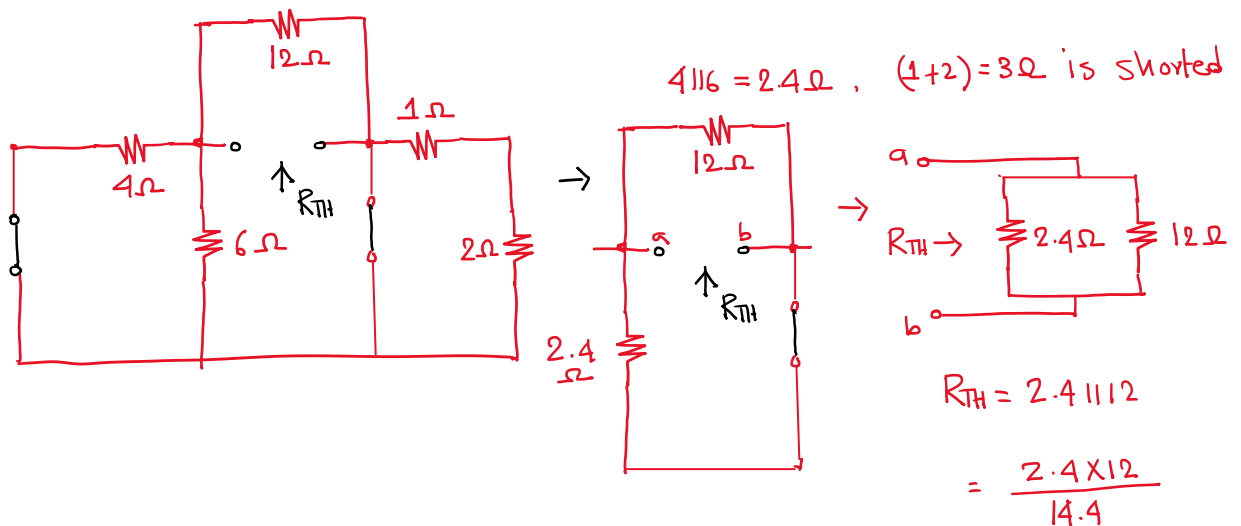
$i_L(\infty) = \frac{24}{4} = 6A$

6Ω Resistor shorted



$$i_L(\infty) = \frac{24}{4} = 6A$$

Determine $\tau = \frac{L}{R_{TH}}$



$$\tau = \frac{L}{R_{TH}} \text{ Sec} = 0.833 \text{ sec}$$

$$= 4.8 \Omega$$

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau} = 6 + [I_b - 6] e^{-t/0.833}$$